AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2025

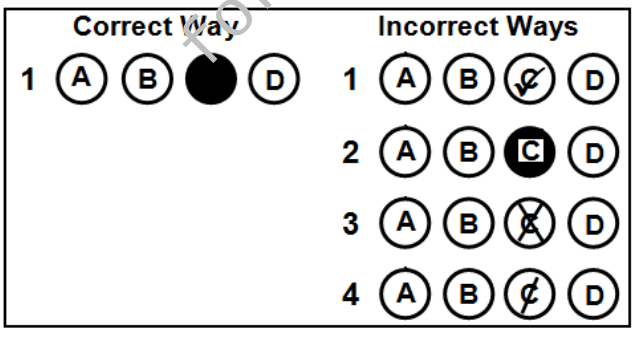
Business Mathematics Paper I

Time: 55 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meanings.

Interest and Annuities

$$I = \frac{PTR}{100} \qquad A = P \times \left(1 + \frac{r}{n}\right)^{nt} \qquad R = \left(1 + \frac{i}{n}\right)^{n} - 1$$

$$j = \left(1 + \frac{r}{m}\right)^{m} - 1 \qquad A = \left(\frac{(1+i)^{n} - 1}{i}\right) \qquad \text{Annuity} = R \times \left(\frac{(1+i)^{n} - 1}{i}\right)$$

Linear Equations, Functions and their Graphs

$$ax + by + c = 0 \qquad \frac{x}{a} + \frac{y}{b} = 1 \quad a \neq 0 \text{ and } b \neq 0$$

$$m = \frac{y_n - y_m}{x_n - x_m} \qquad y - y_m = m(x - x_m)$$

Quadratic Equations, Functions and their Graphs

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad y = a(x-h)^2 + k$$

Derivative of Algebraic Functions

$$\frac{d}{dx}[c] = 0 \qquad \frac{d}{dx} x^n = n \times x^{n-1}$$

$$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \times f'(x) \qquad \frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x) \qquad \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Sequence and Series

$$a_n = a_1 + (n-1)d \qquad S_n = \frac{a_1(1-r^n)}{1-r}, \text{ if } |r| < 1 \qquad S_n = \frac{a_1(r^n - 1)}{r-1}, \text{ if } |r| > 1$$

$$S_n = \frac{n}{2}(2a_1 + (n-1)d) \qquad a_n = a_1 r^{n-1}$$

Matrices and Determinants

$$A^{-1} = \frac{1}{|A|} AdjA \qquad X = A^{-1}B \qquad X = \frac{|A_x|}{|A|}$$

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1. The size of a picture is 4×6 inches. If the picture is to be enlarged in the ratio of 3:2, then the enlarged size will be

   A. 7×8 inches.
   B. 6×9 inches.
   C. 8×18 inches.
   D. 12×12 inches.

2. An annual profit was divided among three partners Asif, Saqib and Nasir. If they received Rs 40,000, Rs 60,000 and Rs 50,000 respectively, then the money was divided in the ratio of

   A. 5:6:4
   B. 4:6:5
   C. 2:5:3
   D. 3:5:2

3. 100 grams of a food item contains 9.3 grams of protein. A serving of 30 grams of this food item will contain how many grams of protein?

   A. 2.79
   B. 2.85
   C. 3.22
   D. 31.0

4. In the new year, Akram's salary was increased from Rs 50,000 to 60,000. The percentage increase in his salary was

   A. 10%.
   B. 11%.
   C. 17%.
   D. 20%.

5. The current value of a certain property in a city is Rs 5,000,000. This includes the 25% increase in its previous value. The value in the previous year was

   A. Rs 1,250,000
   B. Rs 3,750,000
   C. Rs 4,000,000
   D. Rs 6,250,000

6. Zara deposited Rs 5,000 at the end of each period of three months for 6 years at 3% interest compounded quarterly.

   The rate of interest per period is

   A. 0.12
   B. 0.005
   C. 0.0075
   D. 0.00125

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Use the given information to answer Q.7 and Q.8.

Kawish deposited Rs 500,000 in a bank for four years. The bank offers 7% compound interest rate, compounded semi-annually.

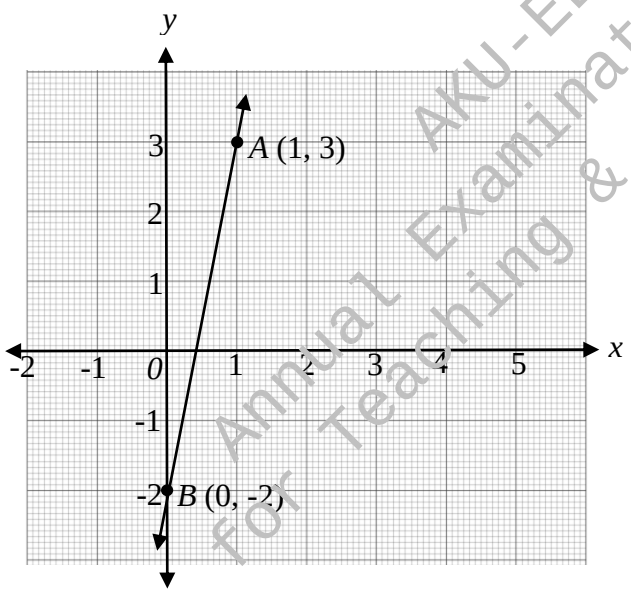
7. The effective rate of interests applicable semi-annually will be

   A. 9.90%.
   B. 7.12%.
   C. 1.31%.
   D. 1.07%.

8. The maximum number of instalment(s) will be

   A. 1
   B. 2
   C. 4
   D. 8

9. The slope of the given line in the graph will be



   A. -5
   B. -1
   C. 1
   D. 5

10. A line has a slope 2 and is passing through the origin (0, 0).

    Which of the following equations would represent the given line?

    A. $x = 2$
    B. $y = 2$
    C. $y = 2x$
    D. $y = x + 2$

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11. If *x*-intercept and *y*-intercept of a straight line are $\frac{1}{2}$ and 3 respectively, then equation of the line would be

A. $6x + y = 3$
B. $x + 6y = 3$
C. $3x + 2y = 6$
D. $2x + 3y = 6$

12. Which of the given pair of equations represents parallel lines?

I. $y = x + 1$
II. $y = 3x + 4$
III. $y = x + 8$
IV. $y = 2x + 4$

A. I and II
B. I and III
C. II and IV
D. III and IV

13. The solution set of the quadratic equation $(x-1)^2 = 1$ will be

A. $\{-1, 1\}$.
B. $\{-1, 0\}$.
C. $\{0, -2\}$.
D. $\{0, 2\}$.

14. What should be added to the expression $36x^2 + 24x$ to make it a perfect square?

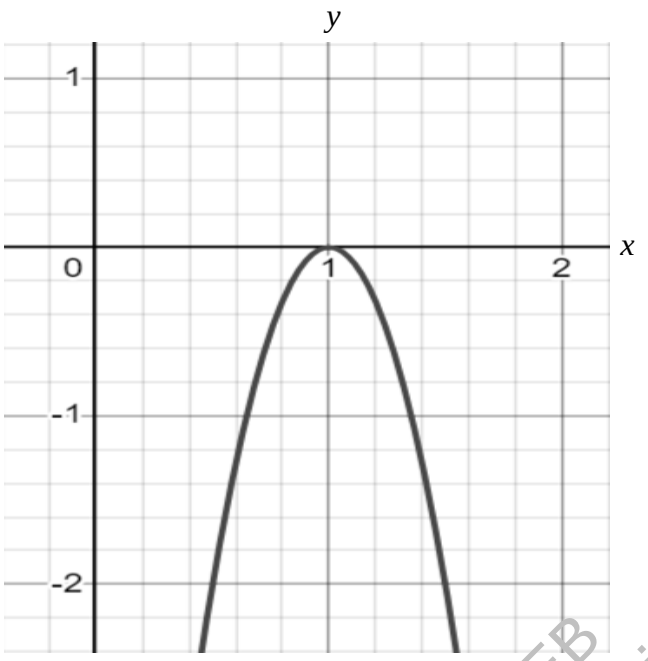
A. 2
B. 4
C. 12
D. 24

15. The vertex of the parabola $y = (x-1)^2 + 1$ will be at

A. $(1, 1)$.
B. $(1, -1)$.
C. $(-1, 1)$.
D. $(-1, -1)$.

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16. The given graph represents a parabola.



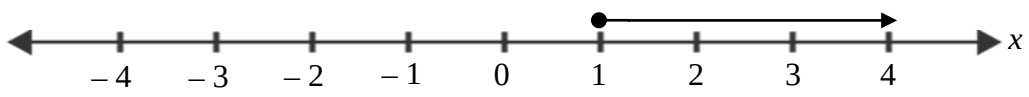
The equation of the given graph will be

A. $y = -8(x-1)^2$
B. $y = 8(x-1)^2$
C. $x = -8(y-1)^2$
D. $x = 8(y-1)^2$

17. The solution set of the simultaneous equations $y = 0$ and $x^2 + y^2 = 9$ will be

A. $\{(-3, 0), (3, 0)\}$.
B. $\{(-3, 0), (3, 3)\}$.
C. $\{(0, -3), (0, 3)\}$.
D. $\{(3, -3), (-3, 0)\}$.

18. The inequality whose solution is represented by the given number line is



A. $x < 1$
B. $x > 1$
C. $x \leq 1$
D. $x \geq 1$

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19. The given table depicts the budget for the payment of utility bills.

Utility	Per Unit Cost (Rs)	Total Budget
Electricity	15	30,000
Gas	40	

If electricity costs Rs x per unit and gas costs Rs y per unit, then this situation will be represented by

A. $15x + 40y \geq 30000$
B. $15x + 40y \leq 30000$
C. $15x - 40y \geq 30000$
D. $15x - 40y \leq 30000$

20. It is given that $M = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$. The additive inverse of M equals to

A. $\begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}$.
B. $\begin{bmatrix} -1 \\ 0 \\ -4 \end{bmatrix}$.
C. $\begin{bmatrix} 1 & 0 & -4 \end{bmatrix}$.
D. $\begin{bmatrix} -1 & 0 & -4 \end{bmatrix}$.

21. Which of the following matrix multiplications is possible?

A. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
B. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a \\ b \end{bmatrix}$
C. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$
D. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b \end{bmatrix}$

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22. The determinant of the matrix $\begin{bmatrix} 5 & 3 \\ 1 & 2 \end{bmatrix}$ is equal to

A. -13
B. -7
C. 7
D. 13

23. If $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & x \end{bmatrix}$ is a singular matrix, then the value of x will be

A. -2
B. -1
C. 1
D. 2

24. The inverse for all the given matrices does not exist EXCEPT

A. $\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}$.
B. $\begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$.
C. $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$.
D. $\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$.

25. The derivative $\frac{d}{dx}\left(a^{10}\right)$, for $a \in R$, will be equal to

A. 0
B. 1
C. $10a^9$
D. $10a^{11}$

26. The derivative of $(2x-1)^2$ with respect to x is

A. $4x-2$
B. $8x-4$
C. $4x-1$
D. $8x-2$

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27. If $g(x) = x(x+2)$, then $g'(x)$ will be

A. 2
B. $2x$
C. $x+2$
D. $2x+2$

28. The slope of the tangent to the curve $y = (x-1)^2$ at (2, 2) will be

A. 1
B. 2
C. 4
D. 6

29. The common ratio of the geometric sequence $\frac{1}{16}, \frac{1}{4}, 4, \ldots$ will be

A. $\frac{1}{16}$
B. $\frac{1}{4}$
C. 1
D. 4

30. In the given arithmetic sequence, 20, 27, 34, 41, …

The 100th term will be

A. 707
B. 713
C. 720
D. 727

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END OF PAPER

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